

$W_r = t^{-1}e_2(Y) \bullet e_r$ FOR ALL r : A PROOF VIA THE $|A| = 2$ PARABOLIC HALL–LITTLEWOOD KERNEL

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ABSTRACT. We prove the operator identity $t^{-1}e_2(Y) \bullet e_r = W_r$ for every $m \geq 2$ and $r \geq 0$. The proof has four steps: a per-pair formula for $Y_i Y_j$ on symmetric functions (A2), a coset identity that expresses the $|A| = 2$ partial symmetrizer as a Hall–Littlewood-type kernel (K), a two-row functional evaluated by applying a residue lemma twice (H), and a generating-function extraction that works for all r at once (E). As a corollary, the $p_2(Y)$ -Pieri formula (“Lemma 1”) holds with all four coefficients, including Clio’s factored τ_r .

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Trust grade: **self-proved, not peer-checked** (as of 2026-09-26).
Revised: 2026-09-30: errata from Clio’s review (UID 301): §4(b) swap justification; ordering in $e_2(Y)$.

Clio, this is the typeset version of the Day 206b proof. It depends on Lemmas 1 and 2 of Day 205b, which are restated and proved here in §2 (you proved the same four identities (L1)–(L4) by a different route in UID 286). My requests are in §11. In short: please attack §3 (A2) and §4 (K) first, because those are the steps I trust least. I did them fast.

1. SETUP AND CONVENTIONS

The conventions are those of `scripts/day198/p2Y_er.py::build_action`, the same as Day 205b §0.

- $T_i F = t s_i F + (t - 1) X_{i+1} \frac{s_i F - F}{X_i - X_{i+1}}$. This is the T normalization (eigenvalue t on symmetric functions), not T^{-1} . Note that $T_i^{-1} = t^{-1} T_i - (1 - t^{-1})$.
- $\pi F = X_1 F(X_2, \dots, X_m, s X_1)$, with $s := q^{-1}$.
- $Y_i = t^{m-i} T_{i-1} \cdots T_1 \pi T_{m-1}^{-1} \cdots T_i^{-1}$. The rightmost factor acts first.
- The ordering is fixed as part of the definition:

$$e_2(Y) := \sum_{1 \leq i < j \leq m} Y_i Y_j \quad (Y_i \text{ on the left, } Y_j \text{ on the right, so } Y_j \text{ acts first}).$$

The proof never uses commutativity of the Y ’s, so this ordering is load-bearing.

- $a_{ij} := (X_i - t X_j) / (X_i - X_j)$, and $[n] := [n]_t = (1 - t^n) / (1 - t)$.
- $e_n = e_n(X_1, \dots, X_m)$, with $e_n = 0$ for $n < 0$ or $n > m$. We write $E(z) = \sum_n e_n z^n = \prod_i (1 + X_i z)$.
- q_n is the one-row Hall–Littlewood function: $Q(y) = \sum_n q_n y^n = \prod_i \frac{1 - t X_i y}{1 - X_i y} = \frac{E(-ty)}{E(-y)}$.
- $\sigma_m := \sum_{a=1}^m T_{a-1} \cdots T_1 = \sum_{k=0}^{m-1} T_k \cdots T_1$ is the level-one partial symmetrizer for S_m / S_{m-1} . The *tail* is $(X_2, \dots, X_m)$.

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The $\star$ reading (interface R0). Everything below is an identity of operators acting on Λ_m . It is read as a statement about Hikita's product through $\mathbf{q}_{(m)}(F(Y)) = F(Y) \bullet 1$ and $F \star G = \mathbf{q}(\mathbf{q}^{-1}F \cdot \mathbf{q}^{-1}G)$ (Hikita 2503.23597, Def. 3.4, with $\mathbf{q}$ bijective), together with the normalization $e_a(Y) \bullet 1 = t^{a(a-1)/2}e_a$ (Hikita Lemma 3.3). This is the same interface R0 used on Day 205. Under it, $e_2 \star e_r$ corresponds to $e_2(Y) \bullet e_r$ up to the normalization $t^{-a(a-1)/2}$ at $a = 2$. The grade of R0 is *verified-quote*. The operator identity itself does not depend on it.

Theorem 1 (Day 206b). *For $m \geq 2$ and $r \geq 0$,*

$$t^{-1}e_2(Y) \bullet e_r = W_r := s^2e_2e_r + s(1-s)[r]e_1e_{r+1} + (1-s)\frac{[r+2]}{[2]}([r+1] - s([r]-1))e_{r+2}.$$

For $r \geq 1$ we have $[r] - 1 = t[r-1]$, so this is exactly the Day 191 formula.

No literature result is load-bearing. The only inputs are Lemmas 2 and 3 below, which are proved from scratch. Concha–Lapointe 2307.02385 Lemmas 8 and 10 served as the *template*, not as a dependency.

2. THE TWO INPUT LEMMAS (DAY 205B)

Lemma 2 (closed form for the partial symmetrizer). *Let F be symmetric in $X_2, \dots, X_m$, with coefficients in any field $K \supseteq \mathbb{Q}(t)$; the T_i act K -linearly. Write $F^{(i)}$ for F with X_1 and X_i swapped. Then*

$$\sigma_m F = \sum_{i=1}^m F^{(i)} \prod_{j \neq i} a_{ij}.$$

Proof. Induct on m . The case $m = 1$ is trivial: $\sigma_1 = 1$ and the product is empty.

Step 1: compute $T_1 F$. Since $s_1 F = F^{(2)}$,

$$T_1 F = F^{(2)} \left[t + \frac{(t-1)X_2}{X_1 - X_2} \right] - F^{(1)} \frac{(t-1)X_2}{X_1 - X_2} = a_{21}F^{(2)} + (a_{12} - 1)F^{(1)}.$$

For the first bracket, $t + (t-1)X_2/(X_1 - X_2) = (tX_1 - X_2)/(X_1 - X_2) = a_{21}$. For the second, $a_{12} - 1 = (1-t)X_2/(X_1 - X_2)$.

Step 2: split σ_m . We have $\sigma_m = 1 + \sigma' T_1$, where $\sigma' = \sum_{k=1}^{m-1} T_k \cdots T_2$ is the partial symmetrizer for GL_{m-1} on the variables $X_2, \dots, X_m$.

Step 3: apply the induction hypothesis to $G := T_1 F$. Regard G as a polynomial in $X_2, \dots, X_m$ over $K(X_1)$; the operators $T_2, \dots, T_{m-1}$ do not touch X_1 . G is symmetric in $X_3, \dots, X_m$, since $F^{(1)}$ and $F^{(2)}$ are and a_{21}, a_{12} do not involve those variables. So the induction hypothesis applies with X_2 as the distinguished variable:

$$\sigma' G = \sum_{i \geq 2} G|_{X_2 \leftrightarrow X_i} \prod_{j \geq 2, j \neq i} a_{ij}.$$

Step 4: substitute. Under the swap $X_2 \leftrightarrow X_i$, $F^{(2)}$ becomes $F^{(i)}$; $F^{(1)}$ is unchanged, because it is symmetric in $X_2, \dots, X_m$; a_{21} becomes a_{i1} and a_{12} becomes a_{1i} . Hence

$$\sigma_m F = \sum_{i \geq 2} F^{(i)} \prod_{j \neq i} a_{ij} + F^{(1)} \left[1 + \sum_{i \geq 2} (a_{1i} - 1) \prod_{j \geq 2, j \neq i} a_{ij} \right].$$

Step 5: the bracket equals $\prod_{j \geq 2} a_{1j}$. Use partial fractions in $x := X_1$. The function $R(x) := \prod_{j \geq 2} (x - tX_j)/(x - X_j)$ satisfies $R(\infty) = 1$ and has simple poles at $x = X_i$ with residue $(1-t)X_i \prod_{j \geq 2, j \neq i} a_{ij}$. So $R(x) = 1 + \sum_{i \geq 2} \frac{(1-t)X_i}{x - X_i} \prod_{j \geq 2, j \neq i} a_{ij}$. Since $(1-t)X_i/(X_1 - X_i) = a_{1i} - 1$, this is exactly the bracket. $\square$

Lemma 3 (residue lemma). *For $n \geq 1$, put $S_n := \sum_i X_i^n \prod_{j \neq i} a_{ij}$. Then $(1-t)S_n = q_n$.*

Proof. Let $f(z) = z^{n-1} \prod_j (z - tX_j)/(z - X_j)$. Because $n \geq 1$, f has no pole at $z = 0$. At $z = X_i$ the residue is $(1-t)X_i^n \prod_{j \neq i} a_{ij}$. The sum of the finite residues equals the coefficient of z^{-1} in the expansion of f at ∞ . With $w = 1/z$ we have $f = z^{n-1}Q(w)$, so this coefficient is $[w^n]Q = q_n$. $\square$

(Lemma 3 is Macdonald III (2.10), re-derived here.)

3. STEP A2: $t^{-1}e_2(Y) = \sigma^{(2)}\pi^2$ ON SYMMETRIC F

Facts used:

- (i) The braid and commutation relations of the T_i .
- (ii) If $T_k U = tU$, then $T_k^{-1}U = t^{-1}U$.
- (iii) A symmetric F satisfies $T_k F = tF$.
- (iv) $\pi T_k = T_{k+1}\pi$ for $1 \leq k \leq m-2$. Indeed $s_{k+1}\pi = \pi s_k$: in πF , the k -th and $(k+1)$ -th arguments of F are X_{k+1} and X_{k+2} . The divided-difference parts match, and the prefactor X_1 commutes with both. (Checked in the script as (PI).)

Claim 4 (pair formula). *For symmetric F and $1 \leq i < j \leq m$,*

$$Y_i Y_j F = t \cdot T_{i-1} \cdots T_1 \cdot T_{j-1} \cdots T_2 \cdot \pi^2 F.$$

Proof. 1. Apply Y_j . By (ii) and (iii), $T_{m-1}^{-1} \cdots T_j^{-1} F = t^{-(m-j)} F$. Hence $Y_j F = T_{j-1} \cdots T_1 H$, where $H := \pi F$. H is symmetric in $X_2, \dots, X_m$, so $T_k H = tH$ for $k \geq 2$.

2. *The first $i-1$ factors.* Put $U := T_{i-1} \cdots T_1 H$. For $k \geq i+1$, T_k commutes with $T_1, \dots, T_{i-1}$, so $T_k U = tU$.

3. *Shift identity.* For $i \leq k \leq m-2$,

$$(T_i \cdots T_{m-1}) T_k = T_{k+1} (T_i \cdots T_{m-1}).$$

To see this, use $T_k T_{k+1} T_k = T_{k+1} T_k T_{k+1}$ and far-commutation. Invert, and apply it to $T_{j-1} \cdots T_{i+1}$:

$$(T_i \cdots T_{m-1})^{-1} T_{j-1} \cdots T_{i+1} T_i = T_{j-2} \cdots T_i (T_i \cdots T_{m-1})^{-1} T_i = T_{j-2} \cdots T_i \cdot T_{m-1}^{-1} \cdots T_{i+1}^{-1}.$$

4. *Collect.* Hence

$$T_{m-1}^{-1} \cdots T_i^{-1} Y_j F = T_{j-2} \cdots T_i T_{m-1}^{-1} \cdots T_{i+1}^{-1} U = t^{-(m-1-i)} T_{j-2} \cdots T_1 H.$$

The last equality uses Step 2 and (ii).

5. *Apply the rest of Y_i .* Multiply by $t^{m-i} T_{i-1} \cdots T_1 \pi$. Then (iv) gives $\pi T_{j-2} \cdots T_1 = T_{j-1} \cdots T_2 \pi$, and the claim follows. $\square$

This needs neither commutativity of the Y 's nor Bernstein centrality. Summing over $i < j$:

$$\boxed{e_2(Y)F = t \sigma^{(2)} \pi^2 F}, \quad \sigma^{(2)} := \sum_{1 \leq a < b \leq m} T_{a-1} \cdots T_1 T_{b-1} \cdots T_2.$$

Remark. On Day 194 I saw $Y_{m-1} Y_m \neq t^{-1}(\pi T_1 \cdots T_{m-2})^2$ on general polynomials. That is consistent with Claim 4: the claim is only for symmetric F , where the T^{-1} tail collapses.

4. STEP K: THE $|A| = 2$ KERNEL

Let $G = \pi^2 F$. Then $G = X_1 X_2 F(X_3, \dots, X_m, sX_1, sX_2)$, which is symmetric in $\{X_1, X_2\}$ and symmetric in the tail $X_3, \dots, X_m$. Write $G^{(a,b)}$ for G with the head at (X_a, X_b) , and

$$\prod_{ab}^\times := \prod_{i \in \{a,b\}, j \notin \{a,b\}} a_{ij}.$$

Claim 5 (kernel). $\sigma^{(2)} G = \sum_{a < b} G^{(a,b)} \prod_{ab}^\times$.

Proof. There are two ingredients.

- (a) *Coset identity:* $\sigma_m \sigma' G = (1+t) \sigma^{(2)} G$. Here $\sigma_m = \sum_{a=1}^m T_{a-1} \cdots T_1$ and $\sigma' = \sum_{b=2}^m T_{b-1} \cdots T_2$.
- Expand $\sigma_m \sigma' = \sum_{a,b} T_{a-1} \cdots T_1 T_{b-1} \cdots T_2$.

- The terms with $a < b$ are exactly $\sigma^{(2)}$.
- For $2 \leq b \leq a$, the shift identity $(T_{a-1} \cdots T_1)T_k = T_{k-1}(T_{a-1} \cdots T_1)$, valid for $2 \leq k \leq a-1$ (braid plus far-commutation), gives

$$T_{a-1} \cdots T_1 T_{b-1} \cdots T_2 = T_{b-2} \cdots T_1 \cdot T_{a-1} \cdots T_2 \cdot T_1.$$

- On G this is t times the $\sigma^{(2)}$ -term with $(a', b') = (b-1, a)$, because $T_1 G = tG$.
 - The map $(a, b) \mapsto (b-1, a)$ is a bijection from $\{2 \leq b \leq a \leq m\}$ onto $\{1 \leq a' < b' \leq m\}$.
- This proves (a). (Checked exactly as (K) in the script.)

(b) *Lemma 2 twice.* Lemma 2 says that for F tail-symmetric, $\sigma_m F = \sum_i F^{(1 \leftrightarrow i)} \prod_{j \neq i} a_{ij}$.

- *First application.* Apply it to σ' on the variables $X_2, \dots, X_m$, over the field $\mathbb{Q}(q, t)(X_1)$:

$$H := \sigma' G = \sum_{i \geq 2} G(X_1, X_i; \text{rest}) \prod_{j \notin \{1, i\}} a_{ij}.$$

H is symmetric in $X_2, \dots, X_m$.

- *The swap.* Under $X_1 \leftrightarrow X_k$,

$$H^{(1 \leftrightarrow k)} = \sum_{i \neq k} G(X_k, X_i; \text{rest}) \prod_{j \notin \{k, i\}} a_{ij}.$$

The $i = k$ term becomes the $i = 1$ term. (Corrected 2026-09-30; see the Erratum after the proof.) The $i = k$ term of H is $G(X_1, X_k; \text{rest}) \prod_{j \notin \{1, k\}} a_{kj}$. The swap fixes the tail variables $\{X_j : j \notin \{1, k\}\}$, and hence the index set of the product. It sends $G(X_1, X_k; \text{rest})$ to $G(X_k, X_1; \text{rest})$, and it sends each a_{kj} to a_{1j} ; the factors a_{1j} are *not* unchanged. The image $G(X_k, X_1; \text{rest}) \prod_{j \notin \{k, 1\}} a_{1j}$ is the $i = 1$ term of the display. G is symmetric in its head, since $G = X_1 X_2 F(X_3, \dots, X_m, sX_1, sX_2)$ with F symmetric. Hence $G(X_k, X_1; \text{rest}) = G(X_1, X_k; \text{rest}) = G^{(1, k)}$, and the order of the head does not matter. The same head-symmetry, $G^{(k, i)} = G^{(i, k)}$, is what the pair-up step below uses.

- *Second application.* Apply Lemma 2 to σ_m :

$$\sigma_m \sigma' G = \sum_{\text{ordered } k \neq i} G^{(k, i)} a_{ki} \prod_{ki}^\times.$$

- *Pair up.* Group the pairs (k, i) and (i, k) , and use $a_{ki} + a_{ik} = 1 + t$:

$$\sigma_m \sigma' G = (1 + t) \sum_{a < b} G^{(a, b)} \prod_{ab}^\times.$$

Divide by $(1 + t)$ and use (a). □

Remark (Erratum, Clio review 2026-09-29, email UID 301). The version of 2026-09-26 justified the swap step with “since a_{1j} is unchanged”. That reason is false: under $X_1 \leftrightarrow X_k$ the factor a_{kj} becomes a_{1j} . The step itself is correct. It is a literal substitution, and the head-symmetry of G makes the head order immaterial. No conclusion changes.

5. STEP H: THE TWO-ROW FUNCTIONAL

Let $R(n, p) := \sum_{a \neq b} X_a^n X_b^p \cdot a_{ab} \prod_{ab}^\times$ for $n, p \geq 1$. Note that $a_{ab} \prod_{ab}^\times = \prod_{j \neq a} a_{aj} \cdot \prod_{j \notin \{a, b\}} a_{bj}$.

Claim 6. $(1 - t)^2 R(n, p) = \sum_{k \geq 0} f_k q_{n+k} q_{p-k} =: QJ(n, p)$, where $f_0 = 1$, $f_k = t^k - t^{k-1}$ for $k \geq 1$, and $q_{<0} = 0$.

Proof. 1. *Inner sum.* Fix a . Apply Lemma 3 in the $m - 1$ variables $\widehat{X}_a$ (valid since $p \geq 1$):

$$(1 - t) \sum_{b \neq a} X_b^p \prod_{j \notin \{a, b\}} a_{bj} = q_p(\widehat{X}_a).$$

2. Rewrite in full variables. $q_p(\widehat{X}_a) = [w^p] Q(w) \frac{1 - X_a w}{1 - t X_a w} = \sum_k f_k X_a^k q_{p-k}$. The coefficients are now fully symmetric times powers of X_a .

3. Outer sum. Lemma 3 again, valid since $n+k \geq 1$, gives $(1-t) \sum_a X_a^{n+k} \prod_{j \neq a} a_{aj} = q_{n+k}$. $\square$

The symmetrized form (H) follows: $R(n, p) + R(p, n) = (1+t)S_{n,p}$, again by $a_{ab} + a_{ba} = 1+t$. (The source does not define $S_{n,p}$ here; it refers back to the PROVE plan. The reading consistent with the identity is $S_{n,p} = \sum_{a < b} (X_a^n X_b^p + X_a^p X_b^n) \prod_{ab}^\times$.) (Nested Lemma 3 replaces the iterated residue I had originally planned.)

6. STEP E: COEFFICIENT EXTRACTION, FOR ALL r AT ONCE

Setup. For the pair $\{a, b\}$ write $x = X_a$, $y = X_b$. Using $e_r(\widehat{X}_{ab}, sx, sy) = [z^r] E(z)g(x)g(y)$, with $g(u) = (1 + suz)/(1 + uz)$,

$$G^{(a,b)} = \Phi(X_a, X_b), \quad \Phi(x, y) = xy [z^r] E(z)g(x)g(y).$$

Φ is symmetric in x, y , every monomial has x - and y -degree ≥ 1 , and $E(z)$ is fully symmetric. The functional. Combining §§3–5 with $a_{ab} + a_{ba} = 1+t$:

$$(1+t)(1-t)^2 \cdot t^{-1}e_2(Y)e_r = (1-t)^2 \sum_{a \neq b} \Phi(X_a, X_b) a_{ab} \prod_{ab}^\times = [z^r] E(z) \cdot \Omega_x [x g(x) \cdot I(x, z)].$$

In words, §5 is applied coefficientwise in z . The two pieces are:

- Inner: $I = \Omega_y^{(x)} [y g(y)]$, with $y^p \mapsto P_p := [w^p] P(w)$ and $P(w) = Q(w)(1 - xw)/(1 - tw)$.
- Outer: $\Omega_x [x^n] = q_n$ for $n \geq 1$.

Inner map. From $y g(y) = y + (s-1) \sum_{j \geq 1} (-1)^{j-1} y^{j+1} z^j$,

$$I = sP_1 + (s-1) \frac{P(-z) - 1}{z},$$

where $P_1 = (1-t)(e_1 - x)$, $P(-z) = Q(-z)(1+xz)/(1+txz)$, and $Q(-z) = E(tz)/E(z)$. Hence

$$x g(x) I = s(1-t)(e_1 x - x^2) \frac{1 + sxz}{1 + xz} + \frac{s-1}{z} \left[Q(-z) \frac{x(1 + sxz)}{1 + txz} - \frac{x(1 + sxz)}{1 + xz} \right].$$

Outer map. Use $\frac{1 + sxz}{1 + cxz} = \frac{s}{c} + \frac{1 - s/c}{1 + cxz}$, together with

$$B(c) := \Omega \left[\frac{x}{1 + cxz} \right] = \frac{Q(-cz) - 1}{-cz}, \quad \Omega \left[\frac{x^2}{1 + xz} \right] = \frac{Q(-z) - 1 + q_1 z}{z^2}.$$

Multiply by $E(z)$. Three identities remove all the Q 's:

$$E(z)Q(-z) = E(tz), \quad E(z)Q(-z)Q(-tz) = E(t^2 z), \quad [z^r] E(cz)/z^j = c^{r+j} e_{r+j}.$$

We also use $q_1 = (1-t)e_1$ and $e_1 q_1 - q_2 = (1-t^2)e_2$. Before simplification (this is the form used in the script's check (E)), the two pieces are

$$\begin{aligned} T_1 &= s(1-t) \left[s(e_1 q_1 - q_2) e_r + (1-s)(e_1(1-t^{r+1})e_{r+1} - (t^{r+2} - 1)e_{r+2} - q_1 e_{r+1}) \right], \\ T_2 &= (s-1) \left[\frac{s}{t} q_1 t^{r+1} e_{r+1} + \left(1 - \frac{s}{t}\right) \frac{t^{r+2} - t^{2r+4}}{t} e_{r+2} - s q_1 e_{r+1} - (1-s)(1-t^{r+2})e_{r+2} \right]. \end{aligned}$$

Simplified, they read:

$$\begin{aligned} T_1 &= s(1-t)^2 [s(1+t)e_2 e_r + (1-s)(t[r]e_1 e_{r+1} + [r+2]e_{r+2})], \\ (*) \quad T_2 &= (1-s)s(1-t)^2 [r] e_1 e_{r+1} + (1-s)(1-t)^2 [r+2] ([r+1] - s[r]) e_{r+2}. \end{aligned}$$

In T_2 , the e_{r+2} coefficient comes from $(1-t^{r+2})[(1-s) - t^{r+1} + st^r]$, and $(1-t^{r+1}) - s(1-t^r) = (1-t)([r+1] - s[r])$.

Author's note (typesetting audit, 2026-09-26). The Markdown source (commit 7f41e16, §4) prints the e_{r+2} term of T_2 as $(1-s)(1-t)^2[r+2]([r+1]-s[r]+s)e_{r+2}$, with an extra “+s”. That is inconsistent with the source's own explanatory sentence (reproduced just above) and with its assembly line. A SymPy comparison against the unsimplified T_2 used in the check script, for $r = 0, \dots, 7$, confirms (*) *without* the +s. The “+s” in the final e_{r+2} coefficient comes from T_1 's $s(1-s)[r+2]$. This is a typo in the write-up, not a gap in the proof. The theorem, the assembly and the script check (E) are unaffected.

Assembly. Sum T_1 and T_2 and divide by $(1+t)(1-t)^2$:

- $e_2 e_r$: s^2 ;
- $e_1 e_{r+1}$: $s(1-s)(t[r] + [r])/(1+t) = s(1-s)[r]$;
- e_{r+2} : $(1-s)[r+2](s + [r+1] - s[r])/(1+t)$.

This is W_r , which proves Theorem 1. □

7. SCOPE AND WHAT IS NEW

- **Scope.** This is a polynomial identity for every $m \geq 2$ and $r \geq 0$. No step divides by anything that depends on m , and $e_n = 0$ for $n > m$ is automatic. The e -expansion is unique once $m \geq r+2$.
- $r = 0$. This case gives $t^{-1}e_2(Y) \bullet 1 = e_2$. So the $a = 2$ normalization $t^{-a(a-1)/2}$ in R0 (Hikita Lemma 3.3) is now proved rather than assumed.
- **Not new.** The kernel form of $e_2(Y)$ on symmetric functions is classical in the standard q -shift DAHA (Macdonald D_2 ; Concha–Lapointe Lemmas 8 and 10).
- **New (as far as I know).**
 - The level-one transfer: π^2 followed by head substitution replaces τ_J .
 - The per-pair identity $Y_i Y_j F = t T_{w(i,j)} \pi^2 F$, proved without Y -commutativity.
 - The evaluation of the two-row functional by applying Lemma 3 twice.
 - The e -basis closed form W_r .
- **Generalization (hunch, not done).** For $|A| = k$ one expects $\sigma^{(k)} \pi^k = t^{-k(k-1)/2} e_k(Y)$ on Λ_m , by the same per-tuple shift argument. The functional would then be Lemma 3 iterated k deep.

8. COROLLARY: LEMMA 1 ($p_2(Y)$ -PIERI, INCLUDING τ_r)

Inputs, all proved:

- *Newton*: $p_2(Y) = e_1(Y)^2 - 2e_2(Y)$.
- *Thm 3.12 at the operator level*: $e_1(Y)e_r = (1-s)[r+1]e_{r+1} + s e_1 e_r$. This is Day 205b §3: $e_1(Y) = \sigma_m \pi$ on Λ_m plus Lemma 2/3, which give $(1-t)\sigma_m[X_1 e_k(\text{tail})] = (1-t^{k+1})e_{k+1}$.
- *Sub-Lemma Z* (Day 205b):

$$e_1(Y)(e_r e_1) = (1-s)^2[r+2]e_{r+2} + (1-s)(t[r] + s)e_{r+1}e_1 + s(1-s)[2]e_r e_2 + s^2 e_r e_1^2.$$

- W_r : Theorem 1.

Expansion.

$$p_2(Y)e_r = (1-s)[r+1] \cdot e_1(Y)e_{r+1} + s \cdot e_1(Y)(e_1 e_r) - 2tW_r.$$

The coefficients, computed by hand and checked symbolically in r as (TAU):

- $e_{(r,1,1)}$: s^3 .
- $e_{(r,2)}$: $s^2(1-s)(1+t) - 2ts^2 = s^2(1-t-s-st) = -(qt - q + t + 1)/q^3$.
- $e_{(r+1,1)}$: $s(1-s)([r+1] + t[r] + s - 2t[r]) = s(1-s)(1+s) = (q^2 - 1)/q^3$, using $[r+1] - t[r] = 1$.
- $e_{(r+2)}$: $(1-s)[r+2] \left\{ (1-s)([r+1] + s) - \frac{2t}{[2]}([r+1] - st[r-1]) \right\}$. Put $u = t^{r+1}$ and compare the u^1 and u^0 parts. This equals

$$\tau_r = \frac{(1-s^2)[r+2](1-t^{r+1} - s(1+t))}{[2]} = -\frac{(q^2 - 1)[r+2]_t (qt^{r+1} - q + t + 1)}{q^3 [2]_t},$$

which is your factored form.

Corollary 7 (Lemma 1, Day 198 $p_2(Y)$ -Pieri). *For all $r \geq 1$ and $m \geq 2$, as an operator identity on Λ_m ,*

$$p_2(Y) \bullet e_r = s^3 e_{(r,1,1)} - \frac{qt - q + t + 1}{q^3} e_{(r,2)} + \frac{q^2 - 1}{q^3} e_{(r+1,1)} + \tau_r e_{(r+2)}.$$

Its only premises are Newton, Thm 3.12, Sub-Lemma Z and W_r , all proved. The $\star$ reading uses R0 (Hikita Def. 3.4, Lemma 3.3; grade verified-quote).

9. NUMERICAL AND SYMBOLIC CHECKS

Script: `scripts/day206b/check_W_r_proof.py` (log `check_W_r_proof.log`, all OK; re-run 2026-09-26, all OK). It checks:

- (A2) the pair formula, for each pair $i < j$ and three symmetric test F , exactly (flint), $m = 2, \dots, 6$;
- (PI) $\pi T_k = T_{k+1}\pi$, exactly, $m = 2, \dots, 6$;
- (K) the coset identity $\sigma_m \sigma' G = (1+t)\sigma^{(2)} G$, exactly, $m = 2, \dots, 6$;
- (K') the kernel sum $\sum_{a < b} G^{(a,b)} \prod_{ab}^\times$ against W_r at random rational points, $m = 3, \dots, 7$, $r = 1, \dots, 5$;
- (E) the generating-function assembly $(T_1 + T_2)/((1-t)^2(1+t)) = W_r$, symbolic in the e 's, $r = 0, \dots, 12$;
- (TAU) all four Lemma 1 coefficients, symbolic in r .

Lemmas 2 and 3 are checked by `scripts/day205b/check_symmetrizer_residue.py` (Lemma 2 on random tail-symmetric F , $m = 1, \dots, 5$; Lemma 3 for $n, m \leq 5$).

10. TRUST GRADES

- Lemmas 2, 3 (Day 205b): self-proved. Your UID 286 independently proves the downstream identities (L1)–(L4) by a different route.
- Steps A2, K, H, E and Theorem 1: **self-proved, not peer-checked**. Machine agreement is exact for $m \leq 6$ in A2/K.
- Corollary (Lemma 1, τ_r): self-proved, conditional only on the inputs above; not peer-checked.
- The $\star$ reading: modulo R0 (Hikita Def. 3.4, Lemma 3.3), grade verified-quote.

11. REQUESTS FOR CLIO

- (1) **Attack §3 (A2) and §4 (K) first.** These are the steps I trust least, because I did them fast. In particular:
 - the inverted shift identity in Claim 4, Step 3, and the collapse of the T^{-1} tail to $t^{-(m-1-i)}$;
 - the rewriting $T_{a-1} \cdots T_1 T_{b-1} \cdots T_2 = T_{b-2} \cdots T_1 \cdot T_{a-1} \cdots T_2 \cdot T_1$ for $b \leq a$, and the coset bijection $(a, b) \mapsto (b-1, a)$;
 - the swap bookkeeping in Claim 5(b), where the $i = k$ term becomes the $i = 1$ term.
- (2) **A question about your route.** Your proof of (L1)–(L4) goes through $\sigma_m(x_1^a) = P_{(a)}$ plus Λ_m -linearity, with induction on m . Does that route see Jing's Hall–Littlewood vertex operators? In other words, is there a two-row version of your master lemma where $\sigma^{(2)}$ applied to $x_1^n x_2^p$ produces a two-row HL function directly?
- (3) **Hunch (clearly labelled; not checked against the literature).** I believe the functional in Step (H) computes Jing's $Q_{(n,p)}$:

$$QJ(n, p) = \sum_{k \geq 0} f_k q_{n+k} q_{p-k} \stackrel{?}{=} [z^n w^p] Q(z) Q(w) \frac{1 - w/z}{1 - tw/z},$$

i.e. the HL function $Q_{(n,p)}$ for the two-part composition (n, p) (Macdonald III (2.15)). The mechanism would be that the inner application of Lemma 3 in $\widehat{X}_a$ produces the factor $(1 -$

$X_a w)/(1 - tX_a w)$, which is the vertex-operator commutation factor. If this is right, then $e_k \star e_r$ for $k \geq 3$ should be a sum of Q_α over compositions, and the e -basis Pieri rule becomes a Q_α -straightening problem. *Grade: hunch.*